UNIT 6 MICROORGANISMS AND GENETIC MODIFICATION

You have seen how some microorganisms are essential decomposers in ecosystems and recycle nutrients, while others are pathogens, causing diseases of animals and plants. There are a few species of microorganisms that are grown by humans in order to make useful products. In this unit we look at this last group, and consider traditional and modern biotechnology. In addition, microorganisms are sometimes genetically engineered to produce new products. This is described in the final chapter, along with some examples of genetically modified plants and animals.



21 USING MICROORGANISMS

In this chapter we look at microorganisms that are grown in order to make products that are of use to humans.

LEARNING OBJECTIVES

- Understand the role of yeast in the production of food, including bread
- Investigate the rate of anaerobic respiration by yeast in different conditions
- Understand the role of bacteria (*Lactobacillus*) in the production of yoghurt
- Understand the use of an industrial fermenter and explain the need to provide suitable conditions for the growth of microorganisms in the fermenter, including aseptic precautions, nutrients, optimum temperature and pH, oxygenation, and agitation.

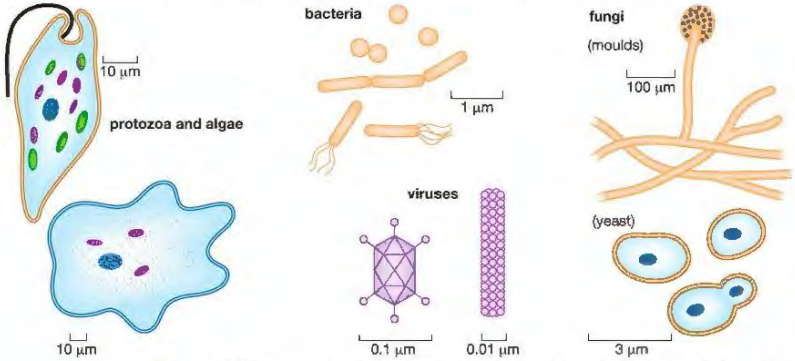
WHAT ARE MICROORGANISMS?

Microorganisms are living things that you can only see with the help of a microscope. The 'bodies' of most microorganisms are made of a single cell, although sometimes millions of cells are gathered together to form a colony. The colony of cells may then be visible to the human eye.

Microorganisms have critical roles to play in recycling the waste products of organisms, as well as recycling the organisms themselves when they die. Many types of microorganisms are studied because they cause disease in animals and plants. On the other hand, humans have made use of the great reproductive capacity of microorganisms to make useful products, such as food, drink and medicines.

There are several groups that we call microorganisms, including protoctists, bacteria, viruses and some fungi (see Chapter 2).

Figure 21.1 shows a few examples of the many types of microorganisms.



▲ Figure 21.1 Some examples of microorganisms. They are not drawn to the same scale. Notice the range of size, as shown by the scale bar alongside each organism. One micrometre (1 μm) is a millionth of a metre, or a thousandth of a millimetre.

FERMENTATION AND BIOTECHNOLOGY

Fermentation and **fermenters** are terms that you will encounter if you are involved in the growth of microorganisms. What do these words mean? Many microorganisms respire anaerobically (see Chapter 1), and originally, fermentation meant any anaerobic respiration process involving microorganisms, such as the fermentation of sugars by yeast. Nowadays, the word is used more generally to define other metabolic processes carried out by microorganisms, many of which are aerobic. Fermentation is normally used to make a useful product.

This overlaps with the definition of **biotechnology**, which means using any organisms (but mainly microbes) to make products that are useful to humans. Although the word itself is relatively new, humans have been using biotechnology processes for thousands of years without knowing it. Since ancient times, fermentation by yeast has been used to make wine and beer and to produce bread. Yoghurt is made by the action of bacteria on milk, and other bacteria and moulds are used in cheese manufacture. Another bacterium is used to convert the ethanol in wine into vinegar.

Our ancestors used biotechnology to make products like these, but they did not understand *how* they were made and had no idea of the existence of microorganisms. Nowadays we understand what is happening when fermentation takes place, and can use biotechnology to produce not just foods but also a huge range of products, from medicines like penicillin to chemicals such as enzymes and fuels.

Modern biotechnology also allows us to alter the genes of microorganisms so that they code for new products. This is called **genetic engineering**. It is a topic that you will read about in Chapter 22.

TRADITIONAL BIOTECHNOLOGY – MAKING FOOD AND DRINKS

When yeast cells are deprived of oxygen, they respire anaerobically, breaking sugar down into ethanol and carbon dioxide:

glucose → ethanol + carbon dioxide

This process has been used for thousands of years to make alcoholic drinks and bread. Both use a species of yeast called *Saccharomyces cerevisiae*.

MAKING ALCOHOLIC DRINKS

Wine is made by using yeast to ferment sugars in grape juice. Commercial wine production takes place in large containers called vats which prevent air reaching the wine and ensure conditions remain anaerobic. Homemade wine is produced in small-scale fermenters fitted with an 'airlock', which allows carbon dioxide to escape but prevents the entry of oxygen (Figure 21.2). The alcohol increases in concentration until it kills the yeast cells, at which point fermentation stops.

Beer is made from barley. Barley contains starch rather than sugars so the starch needs to be broken down first. This happens by allowing the barley seeds to germinate. When they start to germinate they produce the enzyme amylase, which breaks down starch into the sugar maltose. Amylase breaks down starch into the sugar maltose. Later, the maltose from the seeds is fermented by yeast in a large open vat (Figure 21.3).



▲ Figure 21.2 A small fermenter for producing homemade wine. The U-shaped tube at the top is a water-filled airlock that prevents oxygen entering the fermenter but allows carbon dioxide to escape.



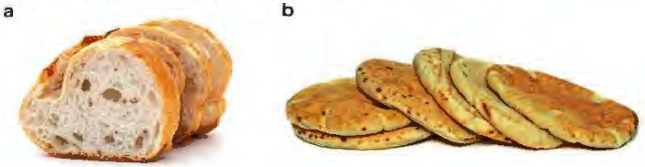
▲ Figure 21.3 A froth forming on the surface of the beer as yeast ferments the sugar to alcohol and carbon dioxide. The carbon dioxide in the froth prevents oxygen entering the mixture from the air – keeping conditions anaerobic.

MAKING BREAD

Yeast is also used to make bread. Wheat flour and water are mixed together and yeast added, forming the bread dough. Enzymes from the original cereal grains break down starch to sugars, which are respired by the yeast. Extra sugar may be added at this stage. In bread-making, the yeast begins by respiring aerobically, producing water and carbon dioxide. The carbon dioxide makes the dough rise. When the air runs out, conditions become anaerobic, so the yeast begins to respire anaerobically making ethanol (alcohol) and more carbon dioxide.

Later, when the dough is baked in the oven, the gas bubbles expand. This gives the bread a light, cellular texture (Figure 21.4). Baking also kills the yeast cells and evaporates any ethanol from the fermentation.

a b



▲ Figure 21.4 (a) The 'holes' in this bread were produced by bubbles of carbon dioxide released from the respiration of the yeast. (b) Bread that is made without yeast is called unleavened bread. What is the difference in texture and appearance between leavened and unleavened bread?

ACTIVITY 1

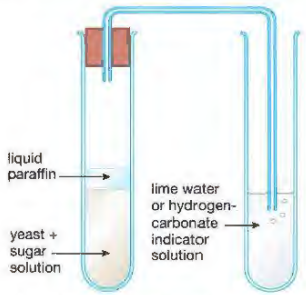
▼ PRACTICAL: INVESTIGATING THE RATE OF ANAEROBIC RESPIRATION IN YEAST

Safety Note: Wear eye protection and avoid skin contact with the indicator.

Some simple apparatus and materials can be used to investigate the rate of anaerobic respiration in yeast.

A small amount of water is gently boiled in a boiling tube to remove any air that is dissolved in the water. The water is allowed to cool, and a small amount of sugar (glucose or sucrose) is dissolved in the water. Finally, a little yeast is added and the mixture is stirred.

The apparatus is set up as shown in Figure 21.5.



▲ Figure 21.5 Apparatus to test for carbon dioxide produced by anaerobic respiration in yeast.

A thin layer of liquid paraffin is added to the surface of the mixture, using a pipette. The boiled water ensures that there is no oxygen in the mixture, and the layer of paraffin stops any oxygen diffusing in from the air. A control apparatus is set up. This is exactly the same as that shown in Figure 21.5, except that boiled (killed) yeast is used instead of living yeast.

Both sets of apparatus are left in a warm place for an hour or two. The mixture with living yeast will be seen to produce gas bubbles. The gas passes through the delivery tube and into the indicator in the second boiling tube.

If this tube contains limewater, it will turn cloudy (milky). If it contains hydrogen carbonate indicator, the indicator will change from orange to yellow. This shows that the gas is carbon dioxide. The time taken for the indicator to change colour is recorded and compared with the control (which will not change).

(If the bung is taken out of the first boiling tube and the liquid paraffin removed using a pipette, the tube will smell of alcohol.)

This method can be used to test predictions, such as:

- the type of sugar (glucose, sucrose, maltose etc.) affects the rate of respiration of the yeast
- the concentration of sugar affects the rate of respiration of the yeast
- how temperature affects the rate of respiration of the yeast.

The rate can be found by timing how quickly the indicator changes colour, or from the rate of production of bubbles of carbon dioxide. You could plan experiments to test these hypotheses.

MAKING YOGHURT

Yoghurt is milk that has been fermented by certain species of bacteria, called **lactic acid bacteria**. The effect of the fermentation is to turn the liquid milk into a semi-solid food with a sour taste.

To make yoghurt, milk is first pasteurised at 85–95 °C for 15–30 minutes, to kill any natural bacteria that it contains, then homogenised to disperse the fat globules. The milk is then cooled to 40–45 °C and inoculated with a starter culture of two species of bacteria, called *Lactobacillus* and *Streptococcus*.

DID YOU KNOW?

Different species of *Lactobacillus* and *Streptococcus* bacteria are also used in the production of cheese.

DID YOU KNOW?

The process called pasteurisation is named after the famous French microbiologist, Louis Pasteur. He was trying to find a way to kill the bacteria in wine that caused it to go sour. In 1864 Pasteur discovered that short periods of heating killed most of the bacteria responsible for this 'spoilage', without affecting the quality of the wine. Today, pasteurisation is used to treat many products, such as milk, fruit juices and canned food.

These bacteria produce lactic acid, as well as starting to digest the milk proteins. The culture is kept at this temperature for several hours while the pH falls to about 4.4 (these are the optimum conditions for the bacteria). The mixture coagulates (thickens) as the drop in pH causes the milk proteins to denature and turn into semi-solids.

When fermentation is finished, the yoghurt is stirred and cooled to 5 °C. Flavourings, colourants and fruit may then be added before it is packaged for sale.

The drop in pH (as the yoghurt forms) gradually reduces the reproduction of the lactic acid bacteria (although it doesn't kill them). It also helps to prevent the growth of other microorganisms, and so preserves the nutrients in the milk. The steps in yoghurt production are summarised in the flow chart (Figure 21.6).

Milk is pasteurised at 85–95 °C for 15–30 minutes.

↓

Milk is homogenised.

↓

Milk is cooled to 40–45 °C and inoculated with a starter culture of lactic acid bacteria.

↓

Mixture is incubated at this temperature for several hours, while bacteria digest milk proteins and ferment lactose to lactic acid.

↓

Thickened yoghurt is stirred and cooled to 5 °C.

↓

Flavourings, colorants and fruit may be added before packaging.

▲ Figure 21.6 Flow chart showing the stages in yoghurt production.

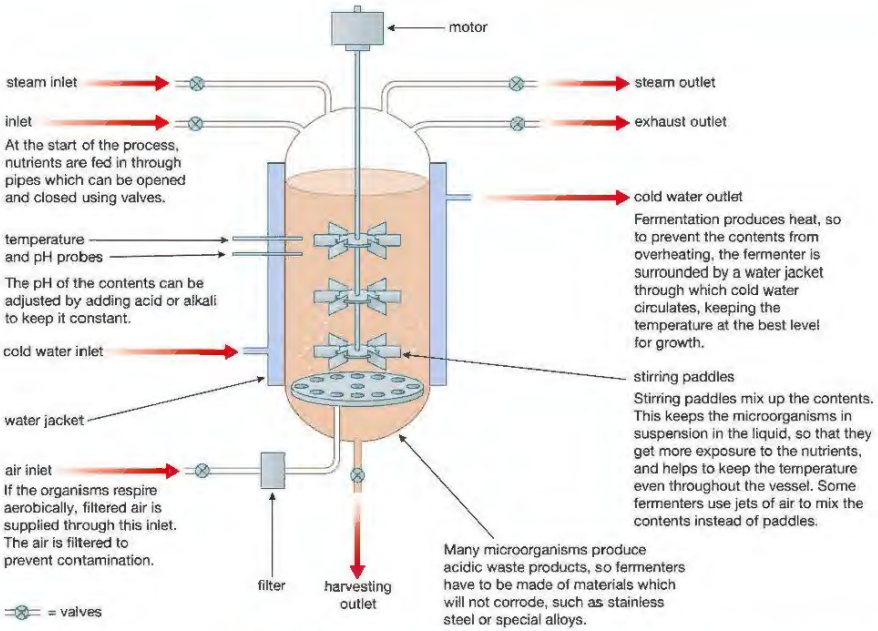
INDUSTRIAL FERMENTERS



◀ Figure 21.7 An industrial fermenter holds hundreds of thousands of dm^3 of a liquid culture.

A **fermenter** is any vessel that is used to grow microorganisms used for fermentation. A glass jar used to make homemade wine is a simple sort of fermenter, even a baking tray containing a ball of dough could be defined as one!

Industrial fermenters are large tanks that can hold up to 200 000 dm^3 of a liquid culture (Figure 21.7). They enable the environmental conditions such as temperature, oxygen and carbon dioxide concentrations, pH and nutrient supply to be carefully controlled so that the microorganisms will yield their product most efficiently. A simplified diagram of the inside of a fermenter is shown in Figure 21.8.



▲ Figure 21.8 An industrial fermenter. Fermenters like this are used to make many products, such as the antibiotic penicillin.

KEY POINT

Many microorganisms use an external food source from their growth medium to obtain energy. In doing this, they change substances in the medium. This is the modern meaning of fermentation. It is used by humans to make many important products. The use of microorganisms to make products useful to humans is called biotechnology.

When fermentation is completed, the products are collected through an outlet pipe. Before the fermenter is filled with new nutrients and culture, the inside of the tank and all the pipes must be cleaned and sterilised. This is usually done with very hot steam under high pressure.

If the inside of the fermenter and the new nutrients are not sterile, two problems are likely to develop. Firstly, any bacteria or fungi that manage to get in would compete with the organism in the culture, reducing the yield of product. Secondly, the product would become contaminated with waste products or cells of the 'foreign' organism. The methods used to prevent contamination by unwanted microorganisms such as filtering the air and sterilising the fermenter using steam, are known as 'aseptic precautions'.

CHAPTER QUESTIONS

More questions on growing useful organisms can be found at the end of Unit 6 on page 301.

SKILLS CRITICAL THINKING

1 It is difficult to make wine containing more than about 15% alcohol. Why is this?

A the alcohol kills the yeast cells
B the yeast cells run out of oxygen
C the yeast cells make too much carbon dioxide
D the yeast cells feed on the alcohol

2 Which of the following is *not* normally controlled in an industrial fermenter?

A Temperature **B** Light intensity **C** pH **D** Oxygen

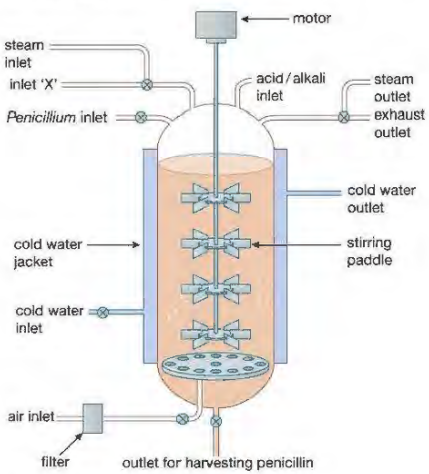
3 During the manufacture of yoghurt, the pH of the milk changes. What causes this change in pH?

A denaturing of the milk proteins
B pasteurisation of the milk
C the conversion of lactose into lactic acid
D yhe action of an enzyme in the milk

4 Yoghurt can be made at home by adding a small amount of old yoghurt to some fresh milk. What is present in the old yoghurt to cause this to happen?

A fat **B** sugar **C** protein **D** bacteria

5 The diagram shows an industrial fermenter which is used to grow the mould *Penicillium* to make the antibiotic penicillin.



SKILLS REASONING

a Explain how the fermenter is sterilised before use.

b Why does air need to be pumped through the fermenter? Why is the air filtered?

SKILLS REASONING (6)

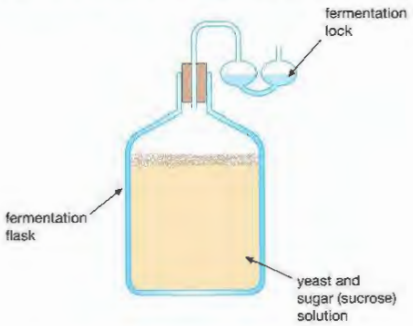
c Explain how a steady temperature is maintained in the fermenter.

d What is supplied through the inlet marked 'X'?

(7) e Suggest what would happen to the growth of *Penicillium* in the fermenter if the paddles stopped working.

SKILLS CRITICAL THINKING (8)

6 The diagram shows a fermentation flask used to make wine at home.



a The yeast cells in the flask respire anaerobically. Write a word equation for this process.

SKILLS REASONING (7)

b Yeast cells can also respire aerobically. How is this prevented in this flask?

c Home winemaking flasks like this are often kept in a warm airing cupboard. Explain why this is done.

SKILLS CRITICAL THINKING (7)

d Wine usually contains no more than about 15% ethanol (alcohol). Explain why.

7 Answer these questions about making yoghurt. Try at first to answer them without looking back to the section on page 285

a Why is the milk pasteurised at the start of the process?

b Why is the mixture of milk and bacteria incubated at 45 °C?

c What causes the milk to thicken?

d Why does fermentation eventually stop?

(8) e Explain how making yoghurt is a way of preserving the nutrients from milk.

22 GENETIC MODIFICATION

In this chapter we will look at ways in which it is now possible to manipulate genes and produce genetically modified organisms – the science of 'genetic engineering'.

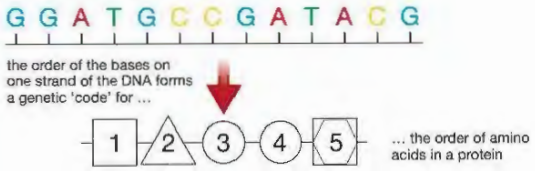
LEARNING OBJECTIVES

- Understand that the term 'transgenic' means the transfer of genetic material from one species to a different species
- Understand how restriction enzymes are used to cut DNA at specific sites and ligase enzymes are used to join pieces of DNA together
- Understand how plasmids and viruses can act as vectors, which take up pieces of DNA, and then insert this recombinant DNA into other cells
- Understand how large amounts of human insulin can be manufactured from genetically modified bacteria that are grown in a fermenter
- Understand how genetically modified plants can be used to improve food production

BIOLOGY ONLY

- Understand how cloned transgenic animals can be used to produce human proteins.

In Chapter 16 you saw that a gene is a section of a molecule of DNA that codes for the production of a protein. The coding strand of the DNA contains triplets of bases, each triplet coding for one amino acid. Different genes produce different proteins because each has a unique sequence of bases that codes for a unique sequence of amino acids (Figure 22.1).



▲ Figure 22.1 The role of DNA in protein synthesis

The protein that is produced could be:

- an enzyme that controls a particular reaction inside a cell or in the digestive system
- a structural protein like keratin in hair, collagen in skin or one of the many proteins found in the membranes of cells
- a protein hormone such as insulin
- a protein with a specific function such as haemoglobin or an antibody.

KEY POINT

A transgenic organism is one that contains a gene that has been introduced from another species. For example, some bacteria have had human genes transferred to them that allow them to make human insulin. Some sheep secrete the protein alpha-1-antitrypsin (AAT) in their milk because they have the human gene that directs the manufacture of this substance. Because they contain 'foreign' genes, they are no longer quite the same organisms. They are transgenic.

RECOMBINANT DNA

The production of **recombinant DNA** is the basis of genetic engineering. A section of DNA – a gene – is cut out of the DNA of one species and inserted into the DNA of another. This new DNA is called 'recombinant' DNA because the DNA from two different organisms has been 'recombined'. The organism that receives the gene from a different species is a **transgenic** organism.